Mid 2023

Q1 (a)

i) $y(n) = 1 + 0.9x(n) + 0.1 \sum_{k=-3}^{3} y(n-k)$

not Linear

$\therefore$ not L.T.I system

ii) $y[n] + 3x[-n-1] = 0$

Linear but not time invariant as there is scale

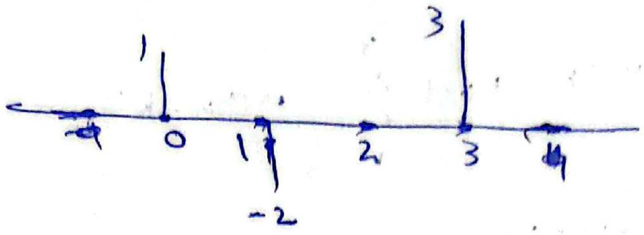
$\therefore$ not L.T.I system

iii) $y(n) = -2y(n-1) + x(n)$

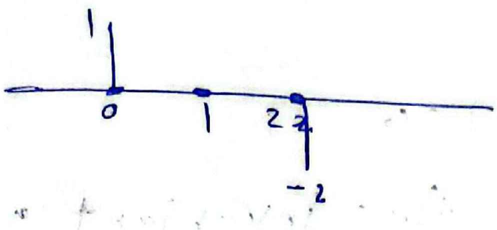
Linear and time invariant

$\therefore$ L.T.I system

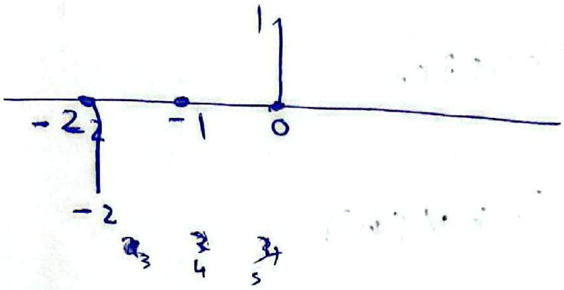
(b)



x(k)



y(k)



y(-k)

i) $z[n] = x[n] * y[n]$

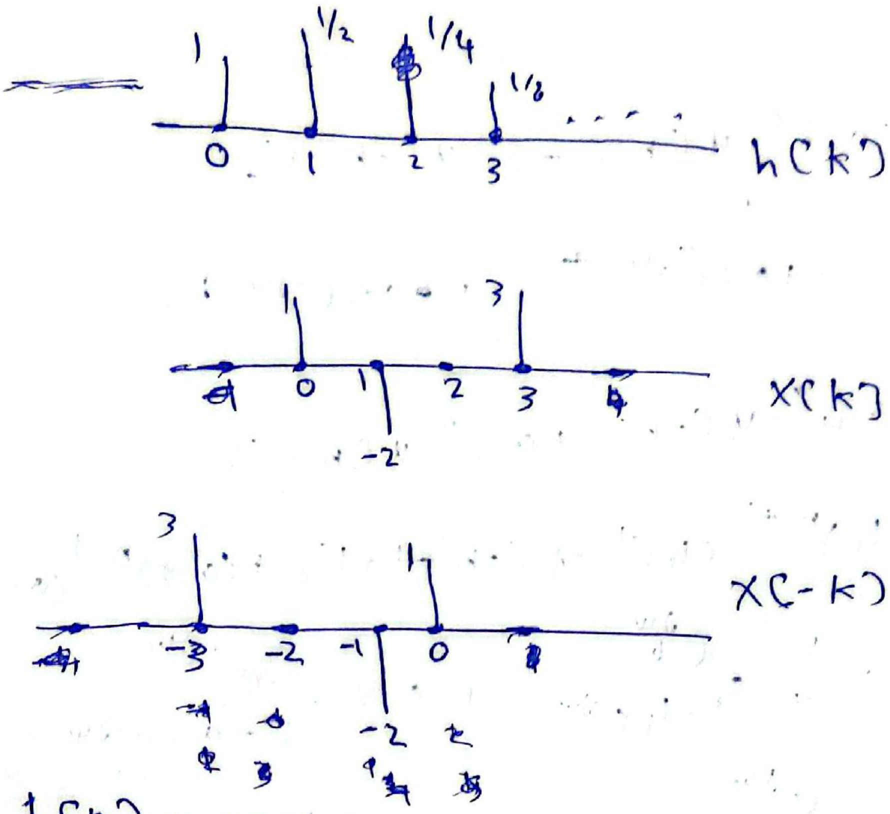
$z[0] = 1$, $z[1] = -2$, $z[2] = -2$, $z[3] = 4 + 3 = 7$

$z[4] = 0$, $z[5] = -6$

$\therefore z[n] = \{1, -2, -2, 7, 0, -6\}$

$\therefore z[n] = \delta[n] - 2\delta[n-1] - 2\delta[n-2] + 7\delta[n-3] - 6\delta[n-5]$

ii)



$y[n] = h[n] * x[n]$

$\therefore\ y[0] = 1 \quad , y[1] = -2 + (\frac{1}{2})$

$y[2] = -2(\frac{1}{2}) + (\frac{1}{4})$

$y[3] = 3 - 2(\frac{1}{4}) + (\frac{1}{8})$

$y[4] = 3(\frac{1}{2}) - 2(\frac{1}{8}) + (\frac{1}{16})$

Q2 (a)

i) $x_1[n] = 0.2^n u[n] - 3^n u[-n-1]$

$$\boxed{X_1(e^{jw}) = \frac{1}{1+0.2e^{-jw}} + \frac{1}{1+0.3e^{-jw}}}$$

ii) $x_2[n] = 4^n u[-n] + 0.3^n u[n-1]$

$$x_2[n] = (4)^{n+1}(4)^{-1} u[-n-1+1] + (0.3)^{n-1}(0.3) u[n-1]$$

$$X_2(e^{jw}) = \frac{1}{4}\frac{e^{jw}}{1-4e^{-jw}} + 0.3\frac{e^{-jw}}{1-0.3e^{-jw}}$$

$$\boxed{X_2(e^{jw}) = \frac{e^{jw}}{4(1-4e^{-jw})} + \frac{0.3e^{-jw}}{1-0.3e^{-jw}}}$$

5

Q_3 (b)

$$H(e^{jw}) = \frac{1}{1-1.5e^{-jw}+0.54e^{-2jw}} = \frac{1}{(1-0.6e^{-jw})(1-0.9e^{-jw})}$$

$$H(e^{jw}) = \frac{A}{1-0.6e^{-jw}} + \frac{B}{1-0.9e^{-jw}}$$

$$A = \frac{1}{1-0.9e^{-jw}}\Bigg|_{e^{-jw}=5/3} \Rightarrow A=-2$$

$$B = \frac{1}{1-0.6e^{-jw}}\Bigg|_{e^{-jw}=10/9} \Rightarrow B=3$$

$$H(e^{jw}) = \frac{-2}{1-0.6e^{-jw}} + \frac{3}{1-0.9e^{-jw}}$$

$$h[n] = -2(0.6)^n u[n] + 3(0.9)^n u[n]$$ i)

$$h[-1] = -2(0.6)^{-1} u[-1] + 3(0.9)^{-1} u[-1]$$

$h[-1]=0 \Rightarrow \therefore h[n]=0 \quad \forall \; n<0$

$\therefore$ h[n] is causal

$$y[n] = x[n] * h[n]$$

$$y[n] = [\delta[n] - \delta[n-1]] [-2(0.6)^n u[n] + 3(0.9)^n u[n]]$$

$$y[n] = -2(0.6)^n u[n] + 3(0.9)^n u[n] + 2(0.6)^{n-1} u[n-1] - 3(0.9)^{n-1} u[n-1]$$

ii)